

Environment & Ecology Core Foundation

Chapter 1: Ecology Principles

Ecosystem Dynamics & Ecological Pyramids

An ecosystem is a functional unit comprising biotic organisms interacting with their abiotic physical environment. Energy flows unidirectionally following Lindeman's 10% Law across trophic levels (Producers → Primary Consumers → Secondary → Tertiary).

Ecological Pyramids: Pyramid of Energy is ALWAYS upright because energy is lost as metabolic heat at every step. Pyramid of Biomass is upright in terrestrial ecosystems but inverted in aquatic ecosystems (standing crop of phytoplankton is smaller than zooplankton). Pyramid of Numbers can be inverted in tree ecosystems supporting numerous parasites.

Ecotone is a transition zone between two distinct biomes (e.g., mangrove between marine and terrestrial), exhibiting high species richness known as the Edge Effect.

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Chapter 2: Conservation Networks

Biodiversity Conservation & Protected Area Network

In-situ conservation protects species in natural habitats: National Parks (highest protection, no grazing allowed, created under Wildlife Protection Act 1972), Wildlife Sanctuaries (limited rights permitted), and Biosphere Reserves (UNESCO MAB program, with Core, Buffer, and Transition zones).

Ex-situ conservation preserves endangered germplasm outside natural habitats: Botanical Gardens, Zoological Parks, Seed Banks, and Cryopreservation facilities.

India hosts 4 global Biodiversity Hotspots (Norman Myers criteria: at least 1500 endemic vascular plant species and lost 70% primary habitat): Western Ghats, Eastern Himalaya, Indo-Burma, and Sundaland (Nicobar Islands).

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Chapter 3: Climate Diplomacy

Climate Change Conventions: UNFCCC, Paris Agreement & Panchamrit

The 1992 Rio Earth Summit birthed the three Rio Conventions: UNFCCC (Climate), CBD (Biodiversity), and UNCCD (Desertification).

The Paris Agreement (COP21, 2015) established the legally binding goal to limit global warming well below 2°C, preferably to 1.5°C above pre-industrial levels, through Nationally Determined Contributions (NDCs) updated every 5 years.

At COP26 Glasgow (2021), India pledged Panchamrit: 1. 500 GW non-fossil energy capacity by 2030; 2. 50% energy requirements from renewables by 2030; 3. Reduce total projected carbon emissions by 1 billion tonnes by 2030; 4. Reduce carbon intensity by 45% by 2030; 5. Achieve Net Zero emissions by 2070.